Dry Lab Pipeline: Strategy and Model Inventory

End-to-End Model from Wound Biology to Voltammogram

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This document answers three questions: what models exist, what type each one is, and how they connect. It is written so any team member can follow the logic, not just the person who built it.

I. The Big Picture

We are building a wound-swab biosensor that reads MMP-9 concentration from a square-wave voltammogram. The dry lab models the full chain from **wound biology** (what concentration of MMP-9 is at the wound site) to **signal readout** (what peak area we measure on a potentiostat).

There are 17 layers (L0–L16). Each layer is a self-contained model that takes inputs from earlier layers and passes outputs forward. The chain is:

Domain	Layers	What it models
Wound biology	L0	MMP-9/HNE concentrations at wound site
Sample handling	L1–L3	Swab collection, dilution, matrix effects
Electrode surface	L4–L12	SAM geometry, probe conformation, ET kinetics
Signal readout	L13–L16	SWV voltammogram, noise, calibration, LOD

The SWV forward model (this document's focus) sits in the signal readout domain. It chains L8 (probe conformation), L10 (ET kinetics), and L13 (SWV simulator).

II. Model Inventory by Type

Ordinary differential equation (ODE) models

ODE models describe how a quantity changes in time. They are the right tool when mixing or reaction happens in a well-stirred compartment.

1. **Layer 0: Wound biology.** Two-compartment ODE for free MMP-9 and HNE in wound fluid, accounting for α2-macroglobulin binding (Olson 1997). Output: $[\text{MMP-9}]_{\text{free}}(t)$. *Source:* `wound_biology.py`
2. **Layer 5: MMP-9 reaction kinetics.** Michaelis-Menten ODE for bulk cleavage of the peptide substrate by MMP-9. This is solution-phase MM; at clinical concentrations (6–30 nM $\gg$ surface probe density $\approx$10 pM) the pseudo-first-order limit applies exactly, so the ODE reduces to $F_k(t) = 1 - e^{-k_{\text{eff}} \cdot [\text{MMP-9}] \cdot t}$. *Source:* `protease_kinetics.py`
3. **Layer 13: BV-SWV simulator.** One surface-confined Butler-Volmer ODE per potential step: $\dot{\Gamma}_{\text{red}} = -k_f \Gamma_{\text{red}} + k_b(1 - \Gamma_{\text{red}})$. Solved analytically in closed form (exponential decay to equilibrium), so it runs as fast as an algebraic expression. 80 distance samples $\times$ N_{steps} potential steps per voltammogram. *Source:* `swv_forward_model.py`

Algebraic / statistical models

These models have no time derivative: they compute a quantity directly from inputs.

1. **Layer 8: Probe conformation (FJC tether).** Freely-jointed chain (FJC) for the SGKIPRTLTA peptide tether. Outputs a half-Gaussian distance distribution $P(d)$ with mean $\langle d \rangle = 21.8$ Å and $\sigma_z = 9.8$ Å. For the PEG6 spacer: Flory radius $R_F = 1.40$ nm sets the hard wall. *Type:* statistical mechanics, no ODE. *Source:* `probe_conformation.py`
2. **Layer 10: Electron transfer kinetics.** Marcus/Butler-Volmer tunnelling: $k_0(d) = k_0^{\text{ref}} \cdot e^{-\beta(d - d_{\text{PEG}})}$. With $\beta = 1.0$ Å^{-1} (Smalley 1995 ★FLAG), $k_{\text{ref}} = 127$ s^{-1} at the PEG6 tip. This is a single algebraic formula, not an ODE. *Source:* `electron_transfer.py`
3. **Layer 15: Calibration.** Linear fit mapping peak area A_{area} to cleavage fraction F_k. Algebraic: $F_k = 1 - A_{\text{area}}/A_{\text{area},0}$. *Source:* `layer15_calibration.py`

What we are NOT using and why

Surface enzyme kinetics (SEK) ODE. A full Michaelis-Menten ODE for surface-confined cleavage would be:

$$\frac{d[S]_{\text{surf}}}{dt} = -\frac{k_{\text{cat}} \cdot [\text{MMP-9}] \cdot [S]_{\text{surf}}}{K_m + [S]_{\text{surf}}}$$

We use pseudo-first-order instead because at clinical concentrations (6–30 nM MMP-9) the enzyme is in at least 600-fold excess over surface probe density (≈10 pM), so $[S] \ll K_m$ and the MM ODE collapses exactly to $dF_k/dt = k_{\text{eff,surf}} \cdot [\text{MMP-9}]$. The SEK ODE would add parameters (k_{cat}, K_m) without changing the answer at any clinically relevant concentration.

PDE / diffusion models. We do not include diffusion of MMP-9 to the electrode surface. At the concentrations and timescales of interest (30 min), diffusion is not rate-limiting: the surface reaction rate is slower than the diffusion supply rate by at least two orders of magnitude (Dauphin-Ducharme 2020, supplementary note 2).

COMSOL thin-layer cell. The Dauphin-Ducharme (DD) COMSOL model includes diffusion and is valid for $\kappa > 12$. Our system operates at $\kappa_{\text{tip}} = 4.2$ (quasi-reversible), well within the surface-confined ODE regime ($\kappa \lesssim 12$). The COMSOL model is validated against our simulator at the regime boundary.

III. How the Models Connect: The Forward Pipeline

The pipeline runs top to bottom. Each row shows one model, its inputs, and what it passes to the next layer.

Layer	Model	Input	Output to next layer
L0	Wound biology ODE	Clinical MMP-9 total	$[\text{MMP-9}]_{\text{free}}(t)$
L1–L3	Sample handling (algebraic)	$[\text{MMP-9}]_{\text{free}}$	[MMP-9] at electrode
L5	Protease kinetics (pseudo-1st)	[MMP-9], time	Cleavage fraction F_k
L8	FJC tether (algebraic)	Peptide sequence	$P(d)$: distance distribution
L10	ET kinetics (algebraic)	$P(d)$, β, d_{PEG}	$k_0(d)$: rate vs distance
L13	BV-SWV ODE (analytic)	F_k, $k_0(d)$, $P(d)$	$\Delta I(E)$: voltammogram
L15	Calibration (algebraic)	$\Delta I(E)$	F_k recovered, LOD

The SWV forward model integrates L8, L10, and L13 into a single runnable script (`swv_forward_model.py`) so that specifying protease concentration and incubation time produces a full predicted voltammogram.

IV. Key Science Decisions

PEG6 spacer, not C6 alkanethiol

The probe uses a PEG6 spacer (6 ethylene oxide units) between the gold surface and the peptide-MB construct. Earlier model versions incorrectly used C6 alkanethiol geometry ($d_{\text{C6}} = 0.756$ nm). The correct hard wall is the PEG6 Flory radius:

$$R_F = N_k^{3/5} \cdot l_{\text{Kuhn}} = 2.76^{0.6} \times 0.76 = 1.40 \text{ nm}$$

This shifts the ET rate from 78,000 s^{-1} (C6 case) to 127 s^{-1} (PEG6 case), moving from the reversible to the quasi-reversible regime: $\kappa_{\text{tip}} = 4.2$. The sensor still works; signal now comes from the peak of $P(d)$ at the PEG6 tip rather than from probes at the mean distance.

MCH backfill: only on the peptide-only electrode

Per lab protocol (Jaden, April 2026): the MCH backfill is present only on the peptide-only electrode (no Fc). The peptide + Fc electrode has no MCH. This distinction matters because MCH changes the surface packing and therefore $P(d)$. The model currently uses the same $P(d)$ for both electrodes; this is a FLAG for experimental refinement.

β for PEG6 is unconfirmed

The tunnelling decay $\beta = 1.0$ Å^{-1} is confirmed for alkanethiol (Smalley 1995). For PEG6 or a peptide bridge, β may differ. Papers to check: Pheeney & Barton (2012) and Smalley et al. (2003), both in `papers/layer_10/`. Until measured, $k_{\text{ref}} = 127$ s^{-1} and $\kappa_{\text{tip}} = 4.2$ carry a FLAG.

V. What Needs Experimental Input

1. **Surface correction factor** η (Layer 15). Ratio of surface to solution cleavage rate. Range 1–100% changes the LOD by 100-fold. Must be measured at Layer 15 by comparing an electrode assay to a solution-phase fluorescence assay on the same batch.
2. **β for PEG6 bridge** (Layer 10). Read Pheeney & Barton 2012 and Smalley 2003. If $\beta_{\text{PEG}} < 1.0$ Å^{-1}, k_{ref} increases and κ_{tip} increases (more reversible, smaller signal).
3. **k_0 by EIS** (trumpet plot). Layer 13 validation. Confirm that $k_0 = 5{,}000$ s^{-1} (from Dauphin-Ducharme for MB-DNA) applies to our MB-peptide-PEG6 system.
4. **$P(d)$ with MCH vs without MCH.** Two different surface geometries need two different $P(d)$ runs. FJC model is the same; the hard-wall boundary may shift if MCH compresses the PEG6 layer.

VI. References

[1] Dauphin-Ducharme et al. (2020). *J. Am. Chem. Soc.* 142, 10,988. `dauphin_ducharme2020.pdf`

[2] Laviron, E. (1979). *J. Electroanal. Chem.* 101, 19. `laviron1979.pdf`

[3] Smalley, J.F. et al. (1995). *J. Phys. Chem.* 99, 13,141. `smalley1995.pdf`

[4] Pheeney, C.G. & Barton, J.K. (2012). *Langmuir* 28, 7,063. `pheeney2012.pdf`

[5] Olson, S.T. et al. (1997). *J. Biol. Chem.* 272, 19,023. `olson1997.pdf`